OXIDATION STATES AND N-FACTOR

08 February 2021 22:18

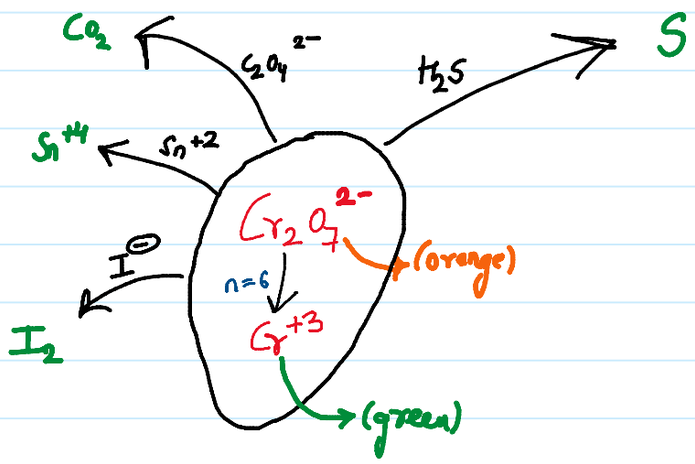
1) $\overset{\ominus}{C}N$ (cyanide) $\quad \overset{+2}{\overset{\ominus}{C}} \equiv \overset{-3}{N}$

2) $\overset{\ominus}{S}CN$ (thiocyanate) $\quad \overset{-2}{\overset{\ominus}{S}} - \overset{+4}{C} \equiv \overset{-3}{N}$

3) $N_3^{\ominus}$ (Azide) $\quad \overset{\ominus}{N} = \overset{\oplus}{N} = \overset{\ominus}{N}$

{ HN_3 → Hydrazoic Acid }

$C_2O_4^{2-} \rightarrow CO_2$; $H_2S \rightarrow S$; $I^{\ominus} \rightarrow I_2$ ($^{-}OH \rightarrow IO_3^-$) ; $Fe^{+2} \rightarrow Fe^{+3}$; $SO_3^{2-} \rightarrow SO_4^{2-}$; $NO_2^{\ominus} \rightarrow NO^{-3}$



K^+ → Purple / Violet / Pink

MnO_4^-

- $\xrightarrow[n=5]{H^+} Mn^{+2}$ (colourless)
- $\xrightarrow[n=1]{\ominus OH}$ (strongly) MnO_4^{2-} (Green)
- $\xrightarrow[n=3]{\text{neutral / feebly Basic}} MnO_2$ (Brown)

* H_2O_2 —acid→ O_2^{2-} + $2H^+$

H_2O_2 —O.A, $n=2$→ H_2O (or) ^-OH ion

H_2O_2 —R.O→ $O_2 + 2e^-$

Disproportionation

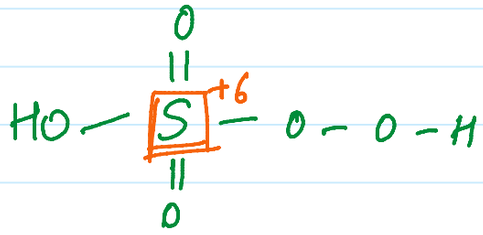
$\overset{0}{Cl_2}$ —Cold dil NaOH→ $\overset{-1}{Cl^-}$ + $^-O\overset{+1}{Cl}$

$\overset{0}{Cl_2}$ —Hot conc NaOH→ $\overset{-1}{Cl^-}$ + $\overset{+5}{ClO_3^-}$

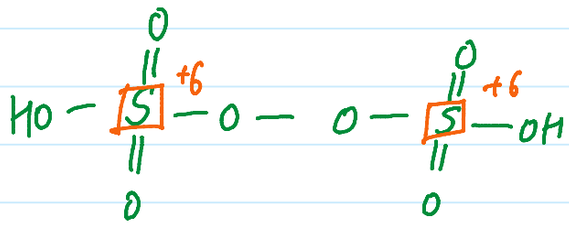
$$\frac{1}{n_{eff}} = \frac{1}{n_o} + \frac{1}{n_R}$$

EXCEPTIONAL D.S

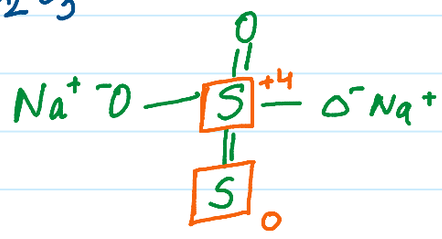
(1) H_2SO_5 CARO's Acid OR Peroxy Mono Sulphuric Acid



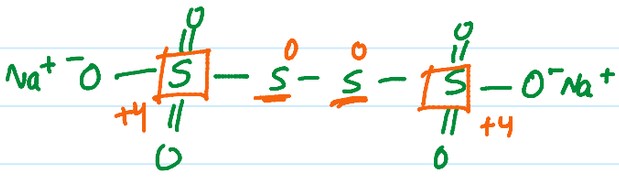
(2) $H_2S_2O_8$ MARSHALL's Acid OR Peroxodi Sulphuric Acid



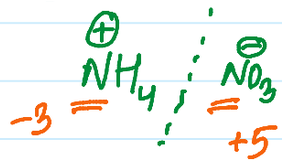
(3) $Na_2S_2O_3$ Sodium Thio Sulphate



(4) $Na_2S_4O_6$ Sodium Theonate



(5) NH_4NO_3 Ammonium Nitrate



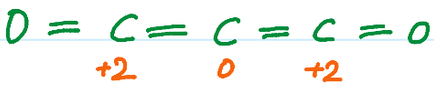
(6) Fe_3O_4 Iron (II) (III) oxide

$FeO \cdot Fe_2O_3$

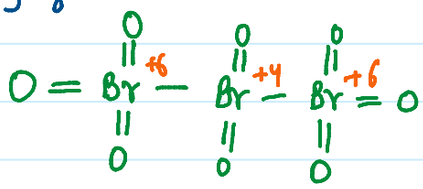
(7) Pb_3O_4

$2PbO \cdot PbO_2$

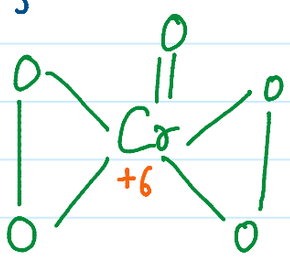
(8) C_3O_2 Suboxide



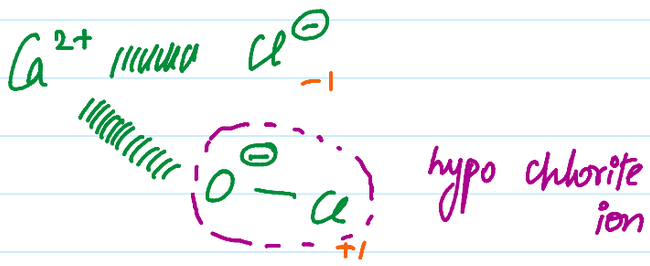
(9) Br_3O_8



(10) CrO_5 Blue Perchromate



(11) $CaOCl_2$ — Calcium Oxy Chloride



hypo chlorite ion

(12) $Na_2[Fe(CN)_5NO]$ — sodium Nitro Prusside

Fe: +2, NO: +1

(13) $Na_4[Fe(CN)_5NOS]$

Fe: +2, NOS: −1

